Akoizo Original Practice - Form 2

Division B | Hovercraft | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) If cushion area doubles and weight stays the same, the required pressure:

A. Stays the same

B. Quadruples

C. Halves

D. Doubles

Answer: _____

2. (2 points) What is the effect of missing impound?

Unit: Impound, inspection, and competition day.
Worked-case data: A team's vehicle passes inspection, but its first attempt stalls behind the start line for 4 seconds. What happens, and what should the team do?
Separate investigation observations:
Inspection: Rear impeller guard has a gap a 3/8-inch dowel fits through.
Team response: Covers the gap with spare mesh from impounded parts.
Re-check: Dowel can no longer reach the impeller.

A. DS, TS, and MS are multiplied by 0.7 for each run

B. Participation points only

C. One run instead of two

D. A 10-point deduction

Answer: _____

3. (2 points) What happens if one loose nickel falls off during a run?

Unit: Carrying nickels: mass, balance, and security.
Worked-case data: A vehicle carries 3 rolls, the half roll, and 12 loose nickels, but 2 loose nickels fall out mid-run. The run is complete. Find the Mass Score and the penalty effect.
Separate investigation observations:
Run 1: 4 rolls in cradles, 20 loose in an open tray: 3 nickels lost at start.
Run 2: Same load, tray with 2 cm walls: nothing lost.
Run 1 score effect: Penalty x 0.8 applied to the whole run.

A. One construction penalty, and that nickel does not count

B. No effect

C. The run is incomplete

D. A penalty for each item lost

Answer: _____

4. (2 points) What average speed is needed for a run time of 18.5 s?

A. 10 cm/s

B. 18.5 cm/s

C. 1 cm/s

D. 3.4 cm/s

Answer: _____

5. (4 points) A run is incomplete with the dowel 74 cm from the finish. Find the Distance Score.

Answer: _____
_____
_____

6. (4 points) Explain the difference between precision and accuracy for run times.

Answer: _____
_____
_____

7. (4 points) How does battery sag affect timing, and how can a team control for it?

Answer: _____
_____
_____

8. (4 points) Give one advantage and one disadvantage of a separate lift fan.

Unit: Levitation: the air cushion and the skirt.
Worked-case data: A hovercraft with nickels weighs 1.6 kg and has a 0.35 m x 0.35 m cushion. What cushion pressure is needed?
Separate investigation observations:
Empty test: Vehicle glides freely across the floor with lift on.
With all nickels: Rear skirt drags and the craft stops quickly.
Nickels moved forward 6 cm: Glides freely again, sitting level.

Answer: _____
_____
_____

9. (4 points) Why is touching the vehicle mid-run so costly?

Unit: Impound, inspection, and competition day.
Worked-case data: A team's vehicle passes inspection, but its first attempt stalls behind the start line for 4 seconds. What happens, and what should the team do?
Separate investigation observations:
Inspection: Rear impeller guard has a gap a 3/8-inch dowel fits through.
Team response: Covers the gap with spare mesh from impounded parts.
Re-check: Dowel can no longer reach the impeller.

Answer: _____
_____
_____

10. (4 points) Why do loose nickels need walls around them?

Unit: Carrying nickels: mass, balance, and security.
Worked-case data: A vehicle carries 3 rolls, the half roll, and 12 loose nickels, but 2 loose nickels fall out mid-run. The run is complete. Find the Mass Score and the penalty effect.
Separate investigation observations:
Run 1: 4 rolls in cradles, 20 loose in an open tray: 3 nickels lost at start.
Run 2: Same load, tray with 2 cm walls: nothing lost.
Run 1 score effect: Penalty x 0.8 applied to the whole run.

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: A hovercraft must cross 185 cm in 12.0 s at steady speed. It enters the timed zone already at speed. What speed is required, and how does adding 0.2 kg affect it?

Answer: _____
_____
_____

12. (2 points) Which plan gives the highest expected score?

Supplied synthetic observations:
Run plan A: No nickels, typical error 0.3 s at TT = 12 s, always completes.
Run plan B: All nickels, typical error 0.6 s, completes 9 times in 10.
Run plan C: All nickels with ramp, error 0.2 s.

A. Plan B

B. Plan A

C. Plan C

Answer: _____

13. (2 points) What setting should the team use for Run 2?

Supplied synthetic observations:
Calibration: Settings 2, 4, 6 gave 8.0, 12.0, 16.0 s.
Target Time: 13.0 s announced.
Run 1 at 4.5: 13.5 s.

A. About 5.0

B. Exactly 6

C. About 4.25

Answer: _____

14. (2 points) Define “Series” in the context of Hovercraft.

Answer: _____
_____
_____

15. (2 points) Define “Footprint” in the context of Hovercraft.

Answer: _____
_____
_____

16. (2 points) Define “Specification check” in the context of Hovercraft.

Answer: _____
_____
_____

17. (2 points) Define “Loose nickel” in the context of Hovercraft.

Answer: _____
_____
_____

18. (2 points) Define “Terminal speed” in the context of Hovercraft.

Answer: _____
_____
_____